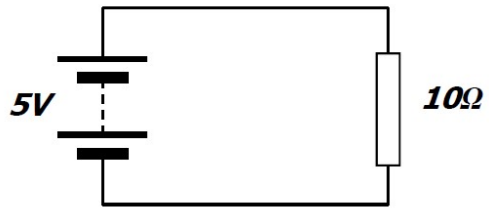
Analogue Electronics Worksheet – Marking Instructions

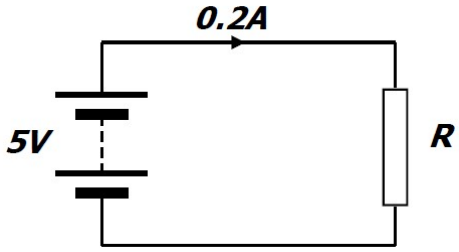
1. Calculate the missing quantity in the following circuits.

   a. Find the current.



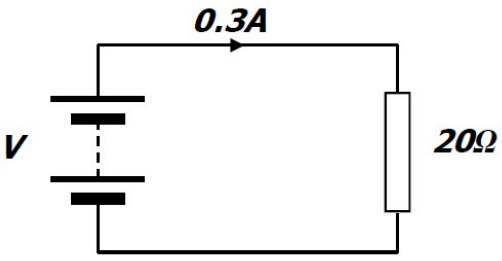
V = IR
I = V / R
I = 5 / 10
I = **0.5A**

   b. Find the resistance of the resistor, R.



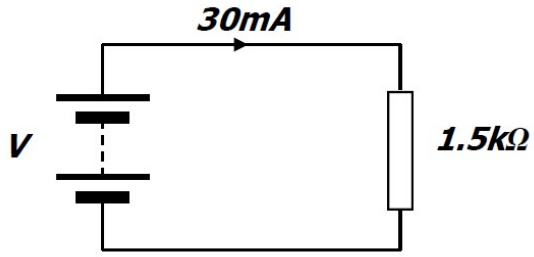
V = IR
R = V / I
R = 5 / 0.2
R = **25Ω**

   c. Find the battery voltage.



V = IR
V = 0.3 x 20
V = **6V**

d. Find the battery voltage.

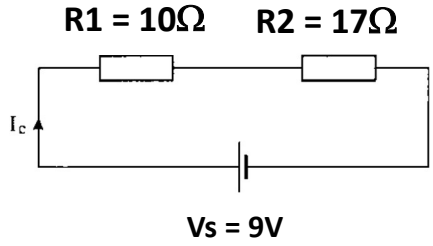


I = 30mA = 30 x 10^{-3} = 0.03A
R = 1.5kΩ = 1.5 x 10^{3} = 1500Ω

V = IR
V = 0.03 x 1500
V = **45V**

2. In your jotter, draw the following series circuits and calculate:
   a) The total resistance (R_T).
   b) The circuit current (I_C). (I = V/R)
   c) The voltage across each resistor. (V = IR)

a)



$R_T = R_1 + R_2$

R_T = 10 + 17 = **27Ω**

I = V / R

I = 9 / 27

I = **0.33A**

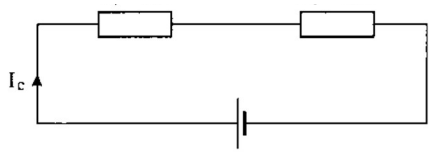
$V_1 = IR_1$

V_1 = 0.33 x 10 = **3.33V**

$V_2 = IR_2$

V_2 = 0.33 x 17 = **5.67V**

b) **R1 = 20Ω** **R2 = 1.5kΩ**



Vs = 12V

$R_T = R_1 + R_2$

$R_T = 20 + 1500 =$ **1520Ω**

$I = V / R$

$I = 12 / 1520$

$I = 0.00789 =$ **0.008A**

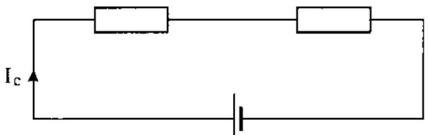
$V_1 = IR_1$

$V_1 = 0.00789 \times 20 = 0.1578 =$ **0.16V**

$V_2 = IR_2$

$V_2 = 0.00789 \times 1500 = 11.835 =$ **11.84V**

c) **R1 = 100Ω** **R2 = 1.3kΩ**



Vs = 12V

$R_T = R_1 + R_2$

$R_T = 100 + 1300 =$ **1400Ω**

$I = V / R$

$I = 12 / 1400$

$I = 0.00857 =$ **0.009A**

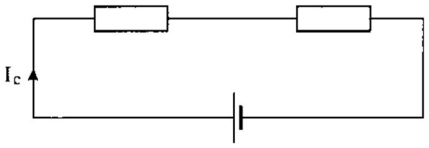
$V_1 = IR_1$

$V_1 = 0.00857 \times 100 = 0.8571 =$ **0.9V**

$V_2 = IR_2$

V_2 = 0. 00857 x 1300 = 11.141 = **11.1V**

d) **R1 = 260kΩ** **R2 = 1MΩ**



Vs = 33 KV

$R_T = R_1 + R_2$

R_T = 260,000 + 1,000,000 = **1,260,000Ω**

I = V / R

I = 33 / 1,260,000

I = 0.000026190 = **0.00003A**

$V_1 = IR_1$

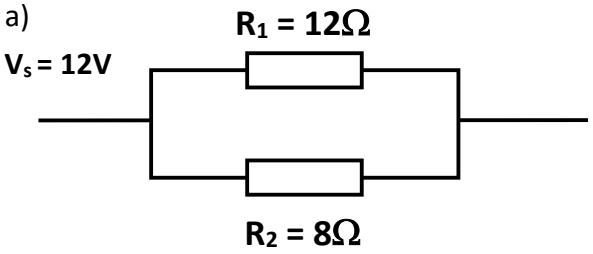
V_1 = 0.000026190 x 260,000= 6.8095238 = **6.8V**

$V_2 = IR_2$

V_2 = 0. 000026190 x 1,000,000 = 26.19 = **26.2V**

3. In your jotter, draw the following parallel circuits and calculate:
   a) The total resistance (R_T).
   b) The circuit current (I_C). ($I_c = V_s/R_T$)
   c) The current across each resistor (I_1 and I_2 and so on).

a)



$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_T = \frac{12 \times 8}{12 + 8} = \frac{96}{20}$$

$$R_T = \mathbf{4.8\Omega}$$

V = IR

I = V / R

I = 12 / 4.8 = **2.5A**

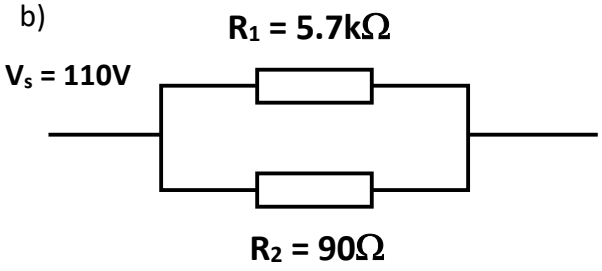
I_1 = V / R_1

I_1 = 12 / 12 = **1A**

I_2 = V / R_2

I_2 = 12 / 8 = **1.5A**

b)



$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_T = \frac{5700 \times 90}{5700 + 90} = \frac{513{,}000}{5790}$$

$$R_T = 88.601\Omega = \mathbf{89\Omega}$$

V = IR

I = V / R

I = 110 / 88.601 = 1.24152 = **1.24A**

I_1 = V / R_1

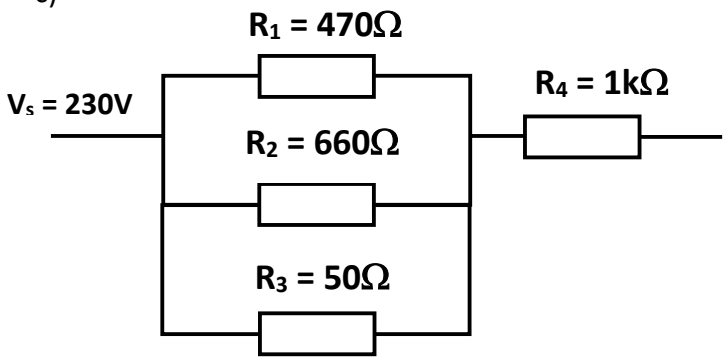
I_1 = 110 / 5700 =0.019298 = **0.02A**

I_2 = V / R_2

I_2 = 110 / 90 =1.2222= **1.22A**

Check, I_{total} = I_1 + I_2 = 0.02 + 1.22 = 1.24A, therefore correct.

c)



Work out parallel resistance first:

$$\frac{1}{R_{Parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{Parallel}} = \frac{1}{470} + \frac{1}{660} + \frac{1}{50} = 0.0236428$$

$$R_{Parallel} = \frac{1}{0.0236428} = 42.296154\Omega$$

Work out series resistance next:

$R_T = R_{Parallel} + R_4$

$R_T = 42.296154 + 1{,}000 = 1042.296154 =$ **1042 Ω**

Work out total current:

$V = IR$

$I_c = V / R$

$I_c = 230 / 1042.296154 = 0.220666 =$ **0.22A**

Work out voltage drop across series part and parallel part of circuit:

$V = I_c R$

$V_{R4} = I_c \times R_4$

$V_{R4} = 0.22 \times 1000 = 220V$

Calculate the Voltage drop across the resistors in parallel. By using Kirchoff's 2nd law we know that the voltage drop across the parallel arrangement must be:

$V_S = V_{R4} + V_{Parallel}$

$V_{Parallel} = V_S - V_{R4}$

$V_{Parallel} = 230 – 220 = 10V$

By using Kirchoff's first law we know that the circuit current I_C will 'split' or divide between the three resistors R_1, R_2 and R_3. We also know that in a parallel circuit the voltage is the same in all branches. The current through R_1 is found by using the resistance of R_1 and the voltage drop across R_1.

$V_{Parallel} = I_1 \times R_1$

$I_1 = V_{Parallel} / R_1$

$I_1 = 10 / 470 = 0.02127 =$ **0.021A**

$V_{Parallel} = I_2 \times R_2$

$I_2 = V_{Parallel} / R_2$

$I_2 = 10 / 660 = 0.0151515 =$ **0.015A**

$V_{Parallel} = I_3 \times R_3$

$I_3 = V_{Parallel} / R_3$

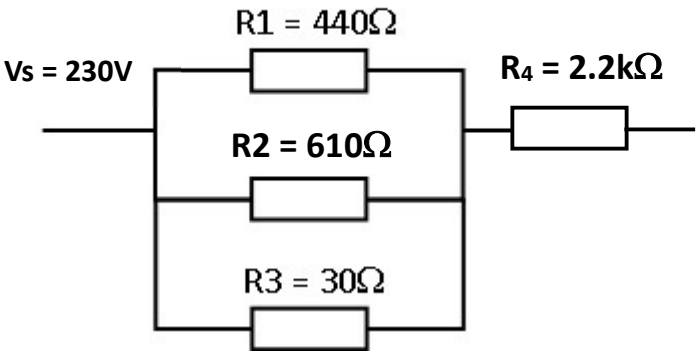
$I_3 = 10 / 50 =$ **0.2A**

Check sum of currents across resistors in parallel equals total current:

$0.021 + 0.015 + 0.2 = 0.236 =$ **0.24A**

Which almost equals I_c **= 0.22A** (error due to rounding)

d)



Work out parallel resistance first:

$$\frac{1}{R_{Parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{Parallel}} = \frac{1}{440} + \frac{1}{610} + \frac{1}{30} = 0.0372454$$

$$R_{Parallel} = \frac{1}{0.0372454} = 26.84894965\Omega$$

Work out series resistance next:

$R_T = R_{Parallel} + R_4$

$R_T = 26.84894965 + 2{,}200 = 2226.84894965\Omega =$ **2,227Ω**

Work out total current:

$V = IR$

$I_c = V / R$

$I_c = 230 / 2226.84894965 = 0.103284958 =$ **0.103A**

Work out voltage drop across series part and parallel part of circuit. Series first as it's easier (if one resistor):

$V = I_c R$

$V_{R4} = I_c \times R_4$

$V_{R4} = 0.103 \times 2200 = 227.2269V$

Calculate the Voltage drop across the resistors in parallel. By using Kirchoff's 2nd law we know that the voltage drop across the parallel arrangement must be:

$V_S = V_{R4} + V_{Parallel}$

$V_{Parallel} = V_S - V_{R4}$

$V_{Parallel} = 230 – 227.2269 = 2.773V$

By using Kirchoff's first law we know that the circuit current I_C will 'split' or divide between the three resistors R_1, R_2 and R_3. We also know that in a parallel circuit the voltage is the same in all branches. The current through R_1 is found by using the resistance of R_1 and the voltage drop across R_1.

$V_{Parallel} = I_1 \times R_1$

$I_1 = V_{Parallel} / R_1$

$I_1 = 2.773 / 440 = 0.00630 =$ **0.0063A**

$V_{Parallel} = I_2 \times R_2$

$I_2 = V_{Parallel} / R_2$

$I_2 = 2.773 / 610 = 0.0045459 =$ **0.0045A**

$V_{Parallel} = I_3 \times R_3$

$I_3 = V_{Parallel} / R_3$

$I_3 = 2.773 / 30 = 0.092433 =$ **0.092A**

Check sum of currents across resistors in parallel equals total current:

$0.0063 + 0.0045 + 0.092 = 0.1028 =$ **0.103A**

Which equals I_c **(0.103A)**

4. Two resistors (15kΩ, 25kΩ) are connected in parallel and then connected to a 6v supply voltage. Calculate;

   a) The total resistance.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_T = \frac{15{,}000 \times 25{,}000}{15{,}000 + 25{,}000} = \frac{375{,}000{,}000}{40{,}000}$$

$$R_T = \mathbf{9375\Omega = 9.375k\Omega}$$

   b) The circuit current.

V = IR

I = V / R

I = 6 / 9375 = **0.00064A**

5. Three resistors (2kΩ, 5kΩ, and 10kΩ) are connected in parallel and then connected to a 10v supply voltage. Calculate;

   a) The total resistance.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{2000} + \frac{1}{5000} + \frac{1}{10000} = 0.00080$$

$$R_T = \frac{1}{0.0008} = \mathbf{1{,}250\ \Omega}$$

   b) The circuit current.

V = IR

I_c = V / R

I_c = 10 / 1,250 = **0.008A**

   c) Current through the 2kΩ resistor.

V = IR

I_{2000} = V / R_{2000}

I_{2000} = 10 / 2000 = **0.005A**

6. A burglar alarm has 2 buzzers connected in parallel, which have a resistance of 200Ω and 300Ω and are connected to an 8v supply voltage. Calculate;

   a) The circuit resistance.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_T = \frac{200 \times 300}{200 + 300} = \frac{60{,}000}{500}$$

$$R_T = \mathbf{120\Omega}$$

   b) The circuit current.

V = IR

I_c = V / R

I_c = 8 / 120 = 0.06666667 = **0.067A**

   c) The current through the 200Ω buzzer.

V = IR

$I_{200\ buzzer}$ = V / $R_{200\ buzzer}$

$I_{200\ buzzer}$ = 8 / 200 = **0.04A**

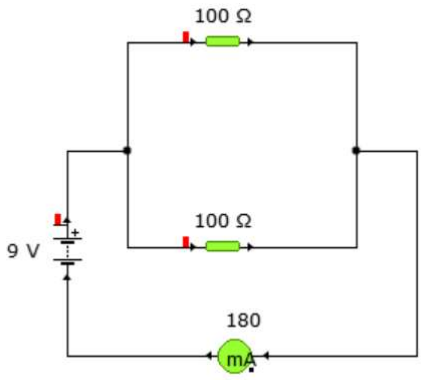
7. A student needs a 50Ω resistance to complete her electronics project. Unfortunately, she only has two 100Ω available. Draw a solution to her problem and explain your answer by providing proof (calculation) that it is correct.

   Place resistors in parallel with each other.

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_T = \frac{100 \times 100}{100 + 100} = \frac{100{,}000}{200}$$

$$R_T = \mathbf{50\Omega}$$



   Checking circuit, V= IR

   R = V / I

   R = 9 / 0.18 = **50Ω**

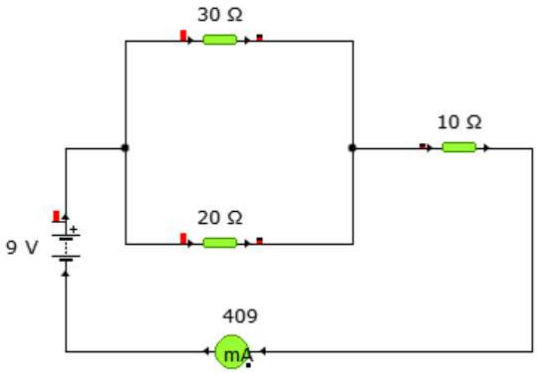
8. A circuit with a voltage supply of 6v has a 30Ω and a 20Ω resistor connected in parallel and then they are connected in series to a 10Ω resistor. Draw the circuit and calculate;

a) The total resistance of the circuit.

$$R_{Parallel} = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_{Parallel} = \frac{30 \times 20}{30 + 20} = \frac{600}{50}$$

$$R_{Parallel} = 12\Omega$$



$R_{Total} = R_{Parallel} + R_{10}$

$R_{Total} = 12 + 10 =$ **22Ω**

b) The circuit current.

$V = IR$

$I_c = V / R$

$I_c = 6 / 22 = 0.2727272727 =$ **0.273A**

c) The voltage dropped over the 10Ω resistor.

$V = I_c R$

$V_{10} = I_c \times R_{10}$

$V_{10} = 0.273 \times 10 = 2.72727272 = 2.73V$

d) The current through the 30Ω resistor.

$V_S = V_{10} + V_{Parallel}$

$V_{Parallel} = V_S - V_{10}$

$V_{Parallel} = 6 - 2.73 = 3.27V$

$V_{Parallel} = I_{30} \times R_{30}$

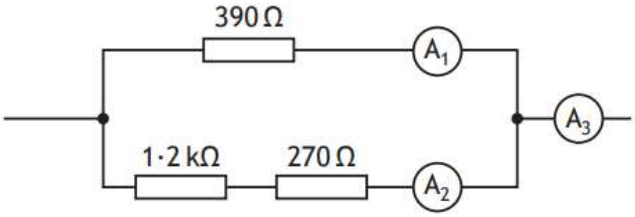
$I_{30} = V_{Parallel} / R_{30}$

$I_{30} = 3.27 / 30 = 0.1090909 =$ **0.109A**

9. **N5 past paper 2019 Q8.** An interactive speaker used in a 'smart home' is shown.



Part of the circuit used in the speaker is shown below.



a) Calculate the total resistance of this circuit 3 marks

$R_{series} = R_2 + R_3 = 1{,}200 + 270 = 1{,}470\Omega$

$$R_T = \frac{R_1 \times R_{series}}{R_1 + R_{series}}$$

$$R_T = \frac{390 \times 1470}{390 + 1470} = \frac{573{,}300}{1860}$$

$R_T = 308.2258 = \mathbf{310\Omega}$ to 2 significant figures

b) The reading on ammeter A_1 is 0.031 A.

(i) Calculate the voltage across the 390 Ω resistor. 2 marks

$V = IR$

$V = 0.031 \times 390 = 12.09 =$ **12V**

(ii) Calculate the current A_2. 3 marks

$V = IR$

$I_{A2} = V / R_{series}$

$I_{A2} = 12 / 1{,}470 = 0.008163 =$ **0.0082A** to 2 significant figures

(iii) Calculate the current A_3. 1 mark

$I_{A3} = I_{A1} + I_{A2}$

$I_{A3} = 0.031 + 0.0082 = 0.03920 =$ **0.039A**